

LoCRANE — Supplementary Material

Longevity-certified residual age network ensemble for transcriptomic aging clocks

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This supplement accompanies the main text. All numbers are read from the frozen result tables released in `results/`; nothing is recomputed after evaluation. The locked protocol is `protocol_lock.json`.

1 Figure source map and band definitions

Every panel of the six result figures is generated from a frozen JSON/NPZ table.

Figure	Source table	Content
Fig. 1 (internal battle)	<code>x41_eq3_from_log.json</code> , <code>x20b...json</code> , <code>x47...json</code> , <code>x48...json</code>	15-split lock, Holm for
Fig. 2 (ext1, GTEx blood)	<code>x45_ext_battle_v4.json</code> , <code>x15c_core3_kshot_v4.json</code>	zero-shot, k -shot, supp
Fig. 3 (ext2, GSE123696)	same as Fig. 2	cross-platform replicat
Fig. 4 (ablation)	<code>x42_ablation.json</code> + screen journal	component costs, six-a
Fig. 5 (attribution)	<code>x43_xai25.json</code> + attribution NPZ	25-seed gated axes
Fig. 6 (mining)	<code>MINING_V4.json</code> + citation audit	levers, modules, geome

Band definitions used in the per-band panels. Age bands are Y (≤ 50), M (51–89), L (≥ 90), following the cohort’s verified longevity structure. In the external figures the age-band panels group each cohort’s samples as out-of-support-young (below 50), in-support (50–89), and, for the microarray cohort, 90+; the support-coverage donuts print the exact arithmetic (32/64 and 27/66 out of the 50–89 training window). The internal per-band radial panel splits the working-age validation range into 50–62 / 63–75 / 76–89 thirds, with radial length proportional to excess MAE.

2 Per-split locked results

Mean absolute error per locked split (working-age validation, $n_{\text{test}} \approx 152$ per split). `eq3'` denotes the shipped hybrid; `base` the unregularized configuration; `pcc` the split-level Pearson correlation of `eq3'`. Paired comparison: 10/15 splits favor `eq3'`, two-sided Wilcoxon $p = 0.035$ (adoption gate,

split	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
<code>eq3'</code>	5.70	5.59	5.71	5.86	5.57	5.58	5.26	6.27	5.14	6.10	5.79	5.52	6.37	6.14	5.52
<code>base</code>	5.68	5.57	5.71	5.84	5.58	5.62	5.28	6.30	5.17	6.08	5.82	5.58	6.42	6.18	5.55

pre-registered).

3 Full composition matrix

All fifteen equal-weight subsets of {deep, EN-40, EN-500, SVR-500} under the locked splits (`x48_composition.json`)
Headline $\text{eq3}' = 5.740$; nearest rival deep+EN-40 at 5.747 (paired $p = 0.36$); single-member configurations significantly worse after Holm correction (deep-only 5.901, EN-40-only 5.839, SVR-500-only 5.898; Holm $p \leq 0.026$); best all-linear combination EN-40+SVR-500 at 5.806.

4 Disclosure ledger

1. **Deep-layer seed direction rate 0.56.** Claims are carried at the ensemble level (10/15, $p = 0.035$); the deep member alone is noisier than the conventional 0.80 bar.
2. **EN-500 is an ensemble member.** The 12/15, $p = 0.013$ comparison is a member-versus-frame comparison; all comparisons against non-member baselines sit at $p \leq 0.006$.
3. **External composition history.** The linear core’s membership was identified with visibility into external few-shot results in an earlier round of the project. The neighboring composition {EN-40, SVR-500} tracks the core within 0.2 years everywhere, and the paper reports all combinations.
4. **Support coverage.** 32/64 (ext1) and 27/66 (ext2) samples lie outside the 50–89 training support; external claims are restricted to relative orderings under identical inputs.
5. **Attribution axis failure.** The FiLM per-gene attribution axis fails its pre-registered split-half gate ($\rho = 0.435 < 0.6$) and is reported as failed; FiLM’s claims rest on its ablation cost and module-level dimorphism spectrum (split-half 0.70).
6. **PFKP grade correction.** PFKP’s mortality association belongs to the source cohort study’s supplementary screening table (HR 2.11, $p = 0.035$), not to its main signature; the paper carries the supplementary-screen grade. The locked replication here is directional (HR 1.20, $p = 0.17$).
7. **Geometry config-dependence.** The class-centroid velocity statistic is reported for the monotonicity-trained representation (mean 0.93, 21/25 seeds below 1); earlier non-regularized configurations showed expansion, so the statistic is tied to the shipped configuration.
8. **Unnominated genes.** ZNF518B and three further claims are graded unsupported by the literature audit and stand as falsifiable model-generated hypotheses.

5 Survival probe details (Vitality 90+, GSE65218)

Cohort: 151 Finnish nonagenarians, Illumina HumanHT-12 v4, 70 deaths over the follow-up window (Jylhävä et al., BMC Medical Genomics 7:54, 2014). Pre-registered hypotheses, fully locked consensus weights: (H1) signature-score Cox model adjusted for frailty and sex — HR 1.14 (95% CI 0.89–1.45, $p = 0.31$), confirmed by 2,000-label permutation; (H2) tertile Kaplan–Meier separation — log-rank $p = 0.47$; (H3) per-gene directional agreement — 7/10 testable genes agree with the model (70%). Nested cross-validated recalibration of the weights reaches C-index 0.540 against 0.585 for covariates alone ($n = 151$; 70 events power only HR ≥ 1.5). Gene-level associations inside the 90+ band: GAL3ST4 HR 1.39 (1.09–1.76, $p = 0.007$); PDZK1IP1 HR 1.21 ($p = 0.095$); PFKP HR 1.20 ($p = 0.17$, directional replication of the supplementary screen); NT5E HR 1.03 ($p = 0.85$,

null). Reading: the longevity signature answers band membership, not survival within the band; the two questions decouple.

6 Demo reproduction

`python code/run_demo.py` runs the full recipe (anchor, FiLM, band auxiliary, monotonicity, hybrid blend) on the released real slice in about two minutes on CPU. Demo-scale numbers verify the pipeline only; the paper’s numbers require the full cohort (Section ?? of the repository README).